AURA Forge Architecture Write-Up

Current evidence packet · 3 tracked Markdown source(s) · 2026-09-01

Final Architecture

Source: docs/capstone/final-architecture.md

System Architecture

```
flowchart LR
    A["Change request"] --> B["Scope and safety precheck"]
    B --> C["Deterministic risk classifier<br/>aura-risk-v1"]
    C --> D["Adaptive router<br/>aura-router-v1"]
    D --> E["LOW route<br/>minimal agents + deterministic checks"]
    D --> F["MEDIUM route<br/>Implementer -> Reviewer -> Tester -> final approval"]
    D --> G["HIGH route<br/>Planner -> approval -> Implementer -> Reviewer -> Tester -> approval -> Project Manager"]
    E --> H["Policy / eval evidence"]
    F --> H
    G --> H
    H --> I["Change Passport + audit trail"]
```

LOW / MEDIUM / HIGH Route Comparison

```
flowchart TB
    L["LOW<br/>docs/non-executable"] --> L1["2 model roles"]
    L1 --> L2["0 human checkpoints"]
    L2 --> L3["16/16 PASS"]

    M["MEDIUM<br/>bounded executable/test"] --> M1["3 model roles"]
    M1 --> M2["1 human checkpoint"]
    M2 --> M3["16/16 PASS"]

    H["HIGH<br/>governance/eval/tool-sensitive"] --> H1["5 model roles"]
    H1 --> H2["2 human checkpoints"]
    H2 --> H3["16/16 PASS"]
```

Evidence Flow / Change Passport

```
flowchart LR
    A["Classifier output"] --> P["Change Passport builder"]
    B["Router output"] --> P
    C["Role sequence and metrics"] --> P
    D["Human approvals"] --> P
    E["Policy/eval results"] --> P
    F["GitHub CI artifacts"] --> P
    P --> G["Versioned JSON passport"]
    G --> H["Grader evidence<br/>docs/capstone/evidence/change-passport-AF-HIGH-001.json"]
```

CI Governance Flow

```
flowchart TB
    A["Push to capstone/aura-forge"] --> B["change-classifier"]
    B --> C["policy-tests<br/>blocking"]
    B --> D["advisory-review<br/>non-blocking"]
    C --> E["evaluation-gate<br/>conditional deterministic eval"]
    E --> F["pipeline-integrity<br/>blocking, if always"]
    D --> G["audit-trail<br/>if always"]
    F --> G
    G --> H["Sanitized artifacts"]
```

What Is Not In The Architecture

- No Render deployment.
- No Vercel deployment.
- No production database write.
- No FitGPT production mutation.
- No real user data.

Governed Semantic Retrieval Evidence

Source: docs/capstone/retrieval-tool-evidence.md

Purpose

The grader reported that semantic document search was absent from the available evidence. AURA Forge now exposes a deterministic semantic-vector retrieval contract through mcp-servers/retrieval/server.py.

This remains a synthetic, course-only corpus. It does not connect to production data, external search, or an embedding API.

Public Tool Contract

Tool: retrieve

Inputs:

- query: required non-empty text.
- requested_classification: optional public, internal, or confidential filter.
- top_k: integer from 1 through 10.

Output schema: governed-retrieval-result-v2

Every returned match includes:

- document ID;
- classification tag;
- text;
- deterministic similarity score;
- citation path and section.

The top-level result also records the role, classification ceiling, normalized query, search mode, returned matches, and withheld matches.

Search Behavior

The implementation converts the query and synthetic documents into deterministic token vectors, expands a small checked synonym map, and ranks with cosine similarity. This is intentionally bounded and reproducible. It does not claim the breadth of a hosted embedding model.

The behavior test uses package audit, which does not appear as an exact phrase in the selected document. The synonym expansion maps it to the dependency, manifest, report, and review terms that rank public-dependency-style first.

Governance Behavior

Authorization runs before search. A role without the retrieval grant receives no result, and an authorized role never receives document text above its classification ceiling. Above-ceiling relevant documents are returned only as withheld metadata with a citation and denial reason.

Policy source: docs/governance-policy.md

Allow-list source: mcp-servers/retrieval/allow-list.json

Verification

```
pytest -q -p no:cacheprovider eval/test_retrieval_behavior.py eval/test_policy.py eval/test_mcp_runtime.py
```

The permanent GitHub policy job includes the retrieval behavior test. The offline container verifier runs with network disabled and a read-only workspace.

Limitation

This proves a governed, schema-validated, classification-tagged, citation-bearing vector retrieval tool over a synthetic corpus. It does not prove production-scale semantic search quality or access to private FitGPT documents.

Reproducibility Runbook

Source: docs/capstone/reproducibility-runbook.md

This runbook lets a grader verify AURA Forge without paid model calls, production access, or secrets.

Fast Path - Under 5 Minutes

From repository root:

```
pytest -q -p no:cacheprovider eval/test_risk_classifier.py eval/test_adaptive_router.py eval/test_pre_aura_control.py
pytest -q -p no:cacheprovider eval/test_ci_change_classifier.py eval/test_pipeline_integrity.py eval/test_audit_trail.py eval/test_c
pytest -q -p no:cacheprovider eval/test_config_docs_consistency.py eval/test_governance_overreach.py
pytest -q -p no:cacheprovider eval/test_retrieval_behavior.py eval/test_reliability_controls.py eval/test_sandbox_contract.py
python3 scripts/check-pipeline-integrity.py .github/workflows/ci.yml
python3 scripts/build-change-passport.py AF-HIGH-001 --output /tmp/aura-passport.json
```

Expected local result from the final packaging run:

```
Run the commands above and use their current output as the result. The historical `82 passed` count is no longer authoritative becau
```

Governed Runtime Path

The trusted local course image is:

```
agentic_engineer_4:latest
```

If needed, rebuild from the trusted repository Dockerfile:

```
docker build -t agentic_engineer_4:latest -f Dockerfile .
```

Run the complete governed verification with a read-only root filesystem, read-only workspace, no network, no Linux capabilities, no credential mount, bounded CPU/memory/processes, and a unique container name:

```
./scripts/run-offline-governance-verification.sh
```

Expected final local result:

```
Current revision result:
```

32 passed

```
The earlier `18 passed` count predates the retrieval, reliability, and sandbox-contract tests.
```

Host Python may not have the MCP package installed. The governed runtime is the authoritative path for MCP/policy verification.

For parallel sessions, each invocation generates a unique container name from UTC time and its process ID. A caller may provide a unique AURA_VERIFY_RUN_ID; duplicate active names fail instead of sharing a container.

The offline verifier needs no Claude authentication volume and performs no paid model call.

Scenario Demonstration Path

Show deterministic classification and routing:

```
PYTHONPATH=eval python3 - <<'PY'
from risk_classifier import classify_change
from adaptive_router import build_route_plan
scenarios = {
    "LOW": ["docs/features/accessibility.md"],
    "MEDIUM": ["web/src/utils/feedbackPrompts.test.js"],
    "HIGH": ["eval/risk_classifier.py", "mcp/coursetools_server.py"],
}
for label, paths in scenarios.items():
    classification = classify_change(paths)
    route = build_route_plan(classification)
    print(label, classification.tier, route.route_id, " -> ".join(route.model_roles))
PY
```

Expected route IDs:

- LOW -> aura-low-v1
- MEDIUM -> aura-medium-v1
- HIGH -> aura-high-v1

Governance Denial Demo

Inspect the real denial:

```
sed -n '1,180p' docs/capstone/governance-overreach-demo.md
pytest -q -p no:cacheprovider eval/test_governance_overreach.py
```

The denial ID is GO-20260811-001.

Escalation / Rollback Check

The adaptive router has a fail-closed escalation test for LOW work that observes a HIGH-sensitive path during implementation:

```
pytest -q -p no:cacheprovider eval/test_adaptive_router.py::test_low_route_observing_high_sensitive_path_fails_closed_with_escalatio
```

Expected result:

```
1 passed
```

This is regression/governance coverage only. It does not change the frozen measured AURA scenario results.

GitHub CI Evidence

Terminal verified run:

https://github.com/Muhammad7839/FitGPT-Agentic-Engineering/actions/runs/31513596822

Expected statuses:

- policy-tests: success
- evaluation-gate: success
- pipeline-integrity: success
- advisory-review: success with SKIPPED - AI SECRET UNAVAILABLE
- audit-trail: success

Safety

No production environment is required. No deterministic gate requires secrets. Advisory AI review safely skips when its optional secret is unavailable. Do not deploy to Render or Vercel for grading.